

Information Visualization Project Overview

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CMSC 734 - Leo Zhicheng Liu

[Video Link](#) || [Webpage Link](#)

I - Summary

Our project is focused on optimizing energy efficiency by identifying buildings that require renovation or conservational measures. It is intended for facility managers at educational institutions, and thus utilizes building-level data from the University of Maryland, College Park. We have built a webpage with an accompanying video walkthrough that the user group can utilize to quickly and efficiently visualize previously unorganized data. It provides a visual context of energy usage intensity (EUI) and the effect of building age and as they relate to carbon dioxide emissions and the trends in energy consumption over time, while also qualifying estimated renovation costs. Providing a historical view on UMD's building emission data enables users to go beyond traditional metrics of emissions intensity and explore the history behind the development decisions that lead to different emissions patterns through the university's history. This will enable users to use UMD's past to inform their decisions on future development and planning. Our team consists of 1) Alex Caramanico, 2) Wooje Chang, 3) Benjamin Knight, and 4) Bahereh Vojdani. The roles were delegated as follows: 1) Tableau visualizations and dashboards, data cleaning, project management, 2) Visualization feedback and refining, scripting and recording of demo video, 3) Javascript and React programming, webpage development and hosting, 4) Webpage description write-ups, data discovery, contextual insights. The following section covers the operative tasks which were used to develop our final system.

II - Tasks

The tasks listed below guided the development of our tool. They are framed as questions and their respective answers are included in the following section so that the effectiveness of the visualizations can be assessed. Each answer is accompanied with annotated screenshots.

1. How does energy consumption vary across building types? Are there any major differences between residential, office, or labs? Which building types are most energy efficient?
2. How does the age of a building contribute to its carbon dioxide emissions and energy efficiency?
3. Are there any outliers as far as energy usage intensity is concerned? Do some buildings require immediate, or speedier attention?
4. Has energy consumption changed in recent years? What improvements or deteriorations have occurred over time?
5. From a cost-based perspective, which buildings are most viable for renovations? In other words, what is the relative value of renovating any particular building?

III - Answers

1. *How does energy consumption vary across building types? Are there any major differences between residential, office, or labs? Which building types are most energy efficient?*

- Besides some notable outliers in the athletics program, labs have by far the most energy consumption on average, while residential buildings have the least. From a gross perspective, the order changes a little bit, but there are very few “School” and “Classroom” buildings. Labs would thus be the highest priority for renovation, while residential buildings the least.

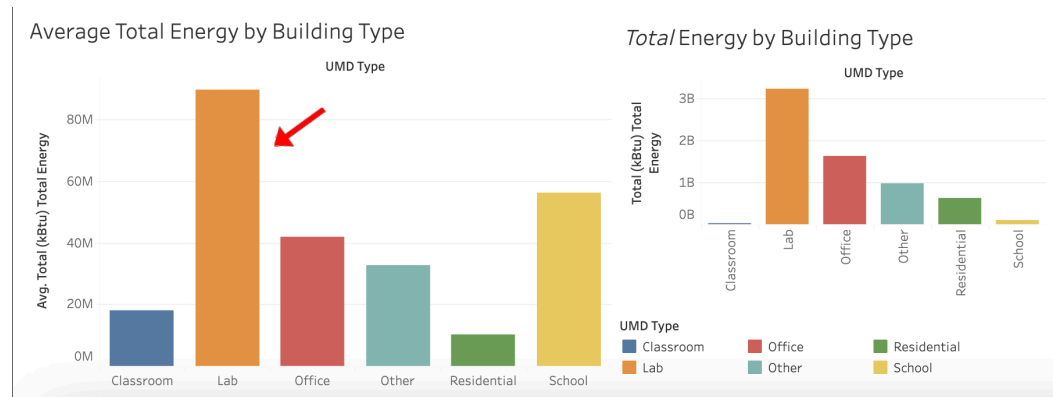


Figure 1: Total energy by building type. Note that Laboratories remain the single most intensive emitter as well as making up nearly half of total UMD emissions.

- EUI difference is perhaps the most important metric from the data. Calculated as the difference between the EPA benchmark EUI and actual EUI, positive values indicate a building is performing well, while negative values suggest the opposite. This provides an intuitive color encoding later on. Looking at the box plots of this metric, a similar conclusion can be made in regards to energy efficiency. Labs are the least efficient, with the lowest median, third quartile, and minimum EUI difference. Residential buildings are fairly well-concentrated around 0 EUI, meaning they are performing relatively well. “Other” and Offices both have major outliers which may require sooner attention, but generally are not performing as inefficiently as Labs.

Energy Performance by Building Type

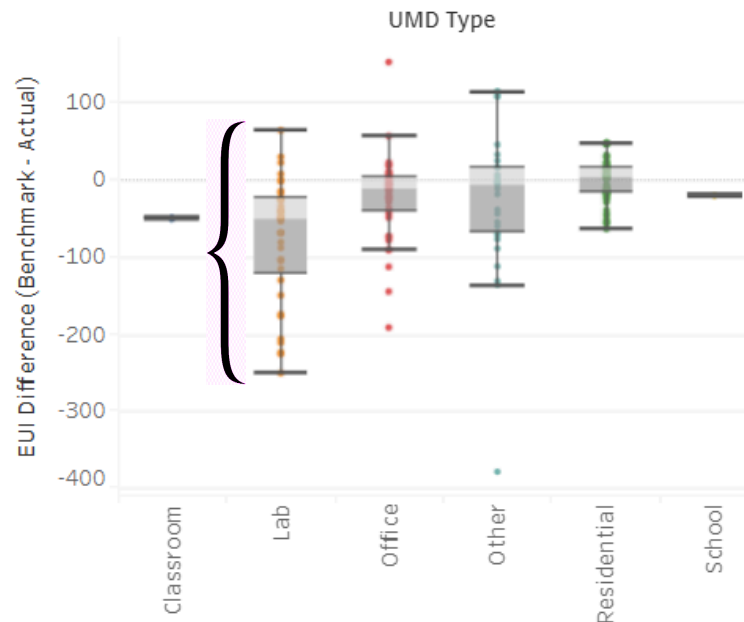


Figure 2: EOI by building type. Note the large outliers in the other, primarily coming from UMD athletics.

2. *How does the age of a building contribute to its carbon dioxide emissions and energy efficiency?*
 - Overall, age is slightly correlated with total carbon dioxide emissions. The older the building, the *less* carbon dioxide emissions it is expected to have. However, there are many outliers and confounds that weaken this relationship. When controlling for each building type, *only residential buildings* have a statistically significant relationship between age and carbon dioxide emissions.

Year Built vs Total CO₂ across Building Types with EUI - ALL

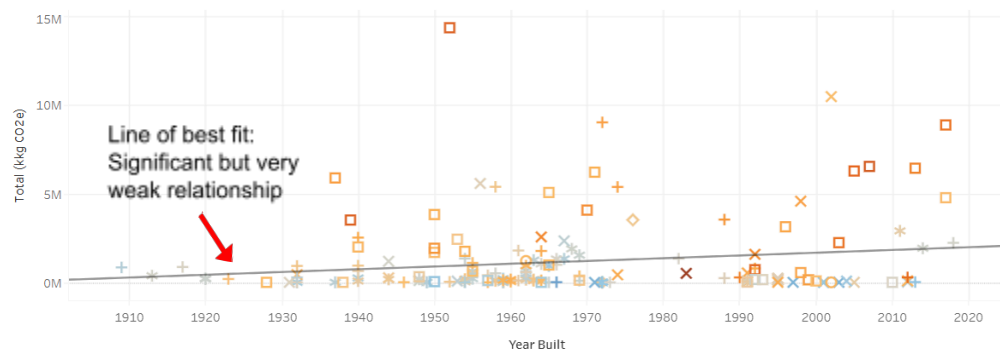


Figure 3: Year built vs Total CO₂. This correlation is weaker as it does not account for differences in building types.

Year Built vs Total CO2 across Building Types with EUI - Residential

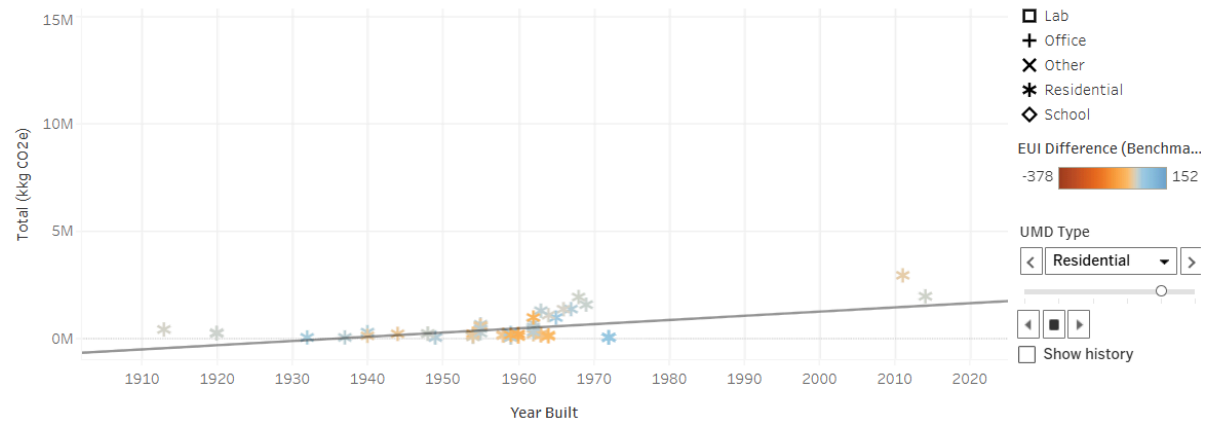


Figure 4: Year build vs. Total CO2 for residential buildings. As the size of housing has grown through UMD’s history, so has its residential emissions.

- In terms of energy efficiency, residential buildings seem to be performing the best according to their benchmark EUI. On the other hand, Labs or “Other” are performing the worst. Offices are performing moderately well. When looking at building age there again seems to be only a loose relationship. Labs built from 1990 to 2007 are generally less efficient than “Other” built in the same time frame. However, labs built from 1948 to 1955 are less efficient than *any* other buildings built in said window. The key takeaway is that these three variables must be controlled for. Total energy is highly correlated to building type, so the type acts as a confound for energy efficiency when factoring in its age.

EUI Performance vs Building Age and Type

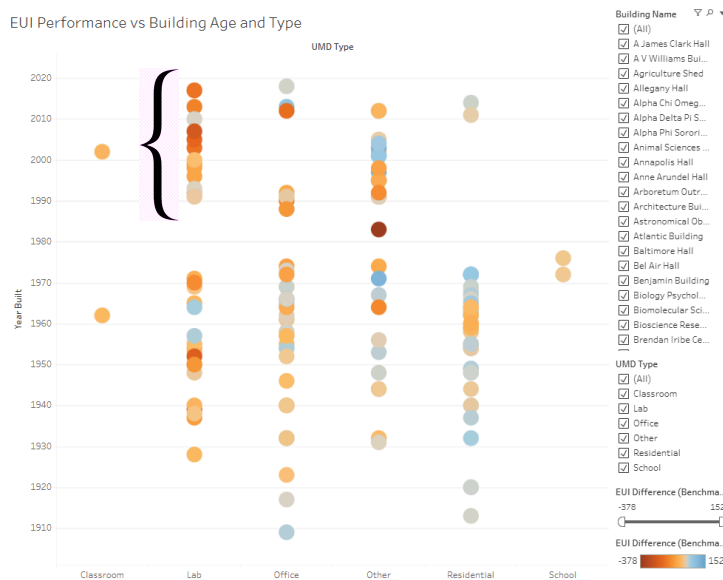


Figure 5: EUI performance vs. Building Age and Type. With this chart type buildings begin to cluster differently based on their type through time.

3. *Are there any outliers as far as energy usage intensity is concerned? Do some buildings require immediate, or speedier attention?*

- The Central Animal Resources Facility is a clear outlier, with by far the lowest EUI difference. Also note the Chemistry Building and the Bioscience Research Building, for the same reason. Almost every other building is pretty tightly packed with others in terms of its energy performance. *(These outliers can be identified in multiple ways from multiple charts).*

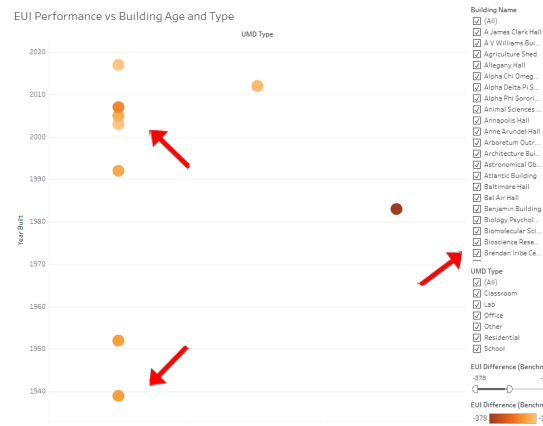


Figure 6: EUI Performance vs. Building Age and Type. Clear outliers in large athletic and campus service projects (e.g. Xfinity Center, Stamp Student Union).

4. *Has energy consumption changed in recent years? What improvements or deteriorations have occurred over time?*

- Using the orange moving average line, it can be seen that total energy consumption has increased after 2020. While this could be due in large part to the COVID-19 pandemic and the construction of newer buildings, both leading to relative increases in campus usage over the last four years, the energy consumption has not returned to its levels at the beginning of 2020. There are often big jumps (see arrows) in monthly energy consumption (blue line) in December and January, as well as June/July. This corresponds to seasons - where more energy is used on heating and air conditioning in the respective months.

Overall Monthly Energy Consumption

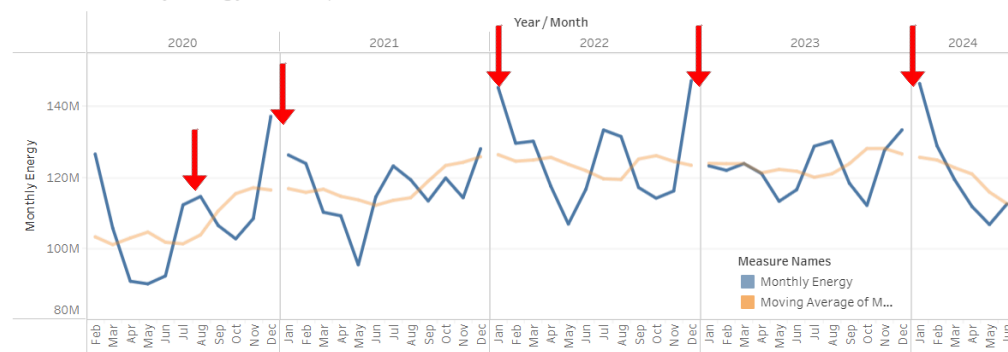


Figure 7: Monthly energy consumption by building. The orange line represents the moving average of monthly energy consumption. Note the peaks during the summer and winter months, independent of semester start/end dates.

- Labs are almost always the greatest average outputters of energy on a monthly basis, while residential buildings are almost always the lowest. **[Figure 8]**
At the building-specific level among Labs, the Chemistry Building has been the greatest outputter for the entire time frame. It can also be seen that the Animal Sciences Agricultural Engineering Building dropped out of the Top 5 among its group in terms of energy consumption - perhaps indicating an improvement. For offices, the Van Munching Hall appears to be very inconsistent - bouncing around in the rankings significantly. For “Other” buildings, the top three are consistently ranked, suggesting little to no improvement. **[Figure 9]**
The color encoding for the building-level line chart shows again the severity of Labs’ poor energy performance, as the entire screen is red. It should be noted that very high energy consumption does not necessarily lead to a poor energy performance metric, as EUI is also a function of building *area*. **[Figure 10]**

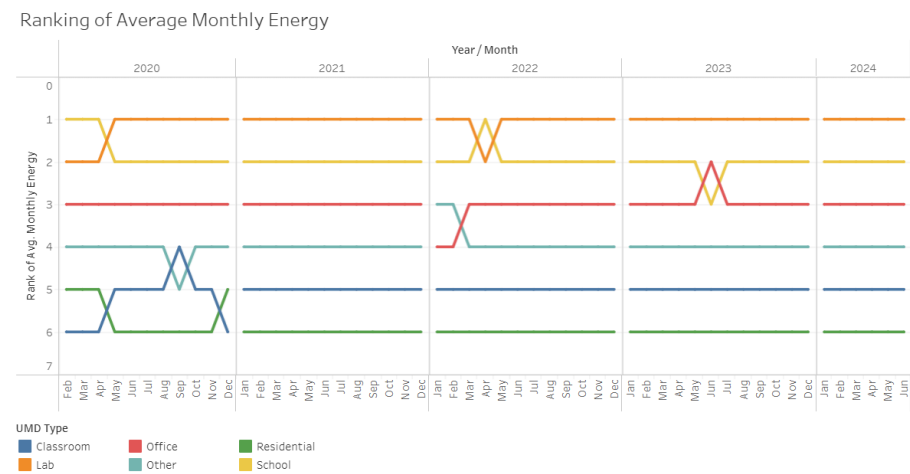


Figure 8: Bump chart ordering greatest to least emissions aggregated by building type throughout the year.

Top 5 Buildings by Monthly Energy (Most) - Lab

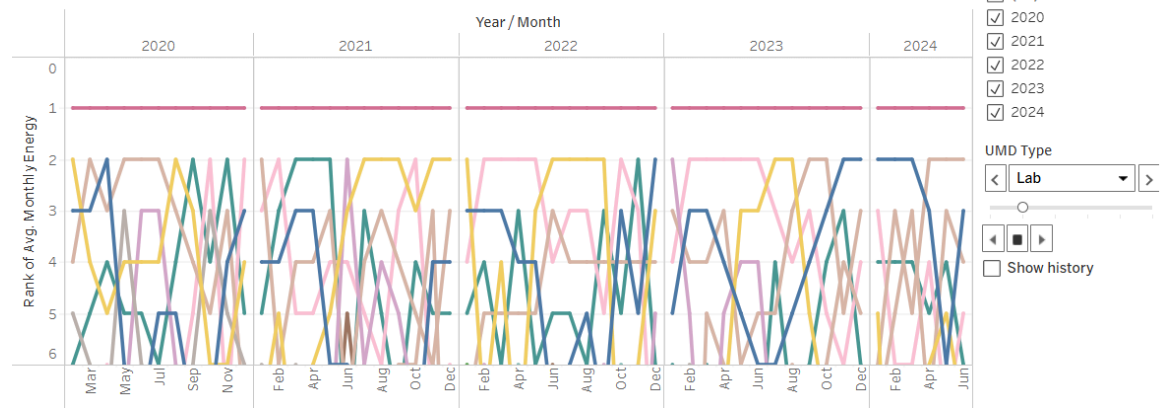


Figure 9: A building specific bump chart of the top emitters through time filtered for only Chemistry buildings. Note that this visualization is interactive which is required to visualize trends in this bump chart.

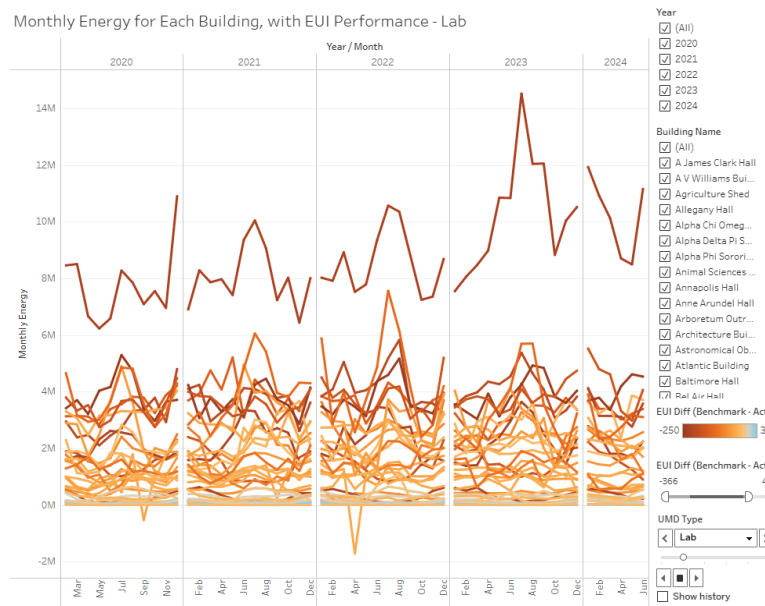


Figure 10: Color-coded monthly energy usage for all lab buildings. All red values mark a negative value, emphasizing greater inefficiencies in UMD’s laboratories.

5. *From a cost-based perspective, which buildings are most viable for renovations? In other words, what is the relative value of renovating any particular building?*
 - Labs and offices would be the most expensive for renovation overhauls [Figure 11]. On the second plot [Figure 12], dots further down and left represent buildings that are energy inefficient and relatively inexpensive to renovate. Disregarding residential buildings due to previous findings, one may point out Central Animal Resources Facility under “Other”, Shuttle Bus Facility under Office, and perhaps Bioscience Research Facility under Lab as viable options for renovation.

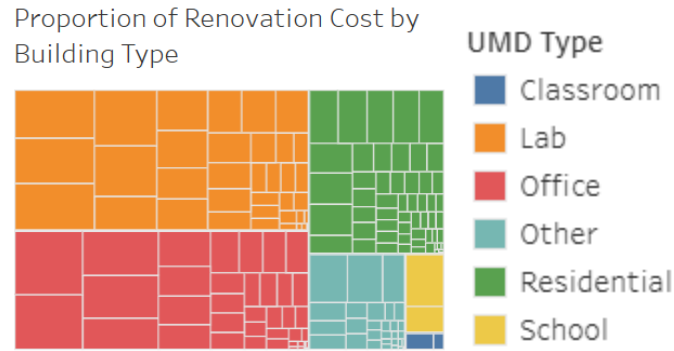


Figure 11: Tree map encoding the relative cost of renovation for each building on the UMD campus, detailed by building type. Note that labs and offices far exceed each of the other categories - some individual buildings accounting for almost 4% of the *combined* renovation cost.

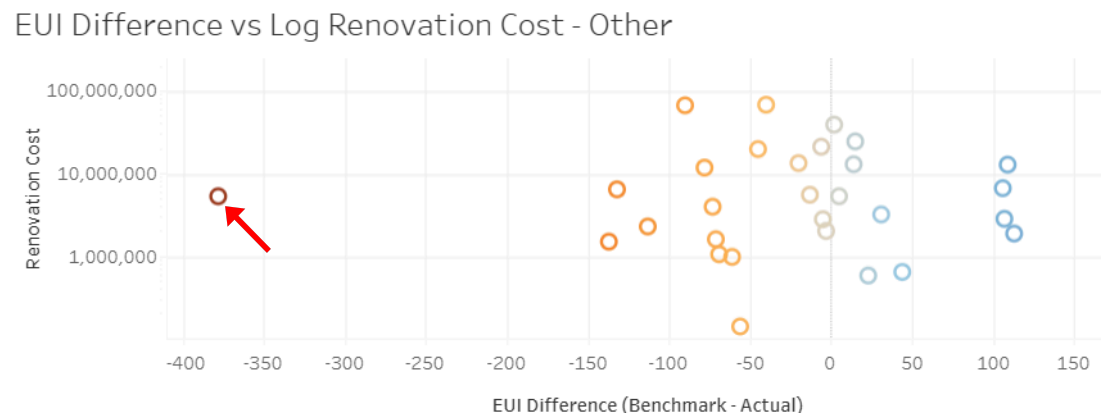


Figure 12: Renovation cost by building compared to EUI. The notable outlier with the arrow is the Central Animal Resources Research Facility.

IV - Uniqueness of Tasks

Our dataset was pulled from UMD's building inventory and public CO2 data. From this data you can access individual building emissions data and a separate dataset with historical emissions. However, the amount of data coupled with issues in formatting prevents users from analysing the data's correlations and implications through time. All of our tasks presented unique challenges in aggregating, cleaning, and visualizing dense and hard to visualize datasets. While there has been work done on emissions planning using this dataset in the past, there has been a noticeable lack of work comparing building types through time and coupling this with an EOI analysis. Our hope is that our dashboard represents a first step in exploring the emissions data across time and will lead to more nuanced visualizations and research connecting this data with other factors such as cost, department square footage, and sub-building type usage groups.